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A Comparative Study of Principal Component Analysis on Different Datasets

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ABSTRACT Principal Component Analysis (PCA) is a sophisticated statistical technique renowned for its ability to distill large datasets into their most essential features. This study evaluates PCA's effectiveness by applying it to both synthetic and real-world datasets, including the Iris and Parkinson's Classification datasets. The methodology involves standardizing the datasets, computing covariance matrices, and determining principal components through eigenvalues and eigenvectors. Results demonstrate that PCA efficiently reduces dimensionality, facilitating data visualization and interpretation. Nonetheless, PCA's limitations, such as sensitivity to outliers and assumptions of linearity, are acknowledged. Despite these challenges, PCA remains a crucial tool for uncovering insights from complex datasets.

INDEX TERMS Principal Component Analysis, Dimensionality Reduction, Data Visualization, Eigenvalues, Feature Extraction.

I. INTRODUCTION

Principal Component Analysis (PCA) is a widely used technique that is often used for reducing the dimensionality of large datasets while retaining most of the original information. PCA transforms high-dimensional dataset into a smaller set of uncorrelated variables, known as principal components, which simplifies the processes of data exploration, visualization, and analysis. PCA effectively identifies crucial features, reduces representation error, and uncovers underlying simplified structures. This enhances data interpretability and decreases computational demands. PCA also minimizes, or altogether eliminates, common issues such as multicollinearity and overfitting. PCA is a commonly used approach within regression analysis but it is also leveraged for a variety of use cases, such as pattern recognition, signal processing, image processing, and more. Mastering PCA provides a strong foundation for more advanced dimensionality reduction techniques and equips

learners with the skills needed to address complex data analysis tasks.

II. RELATED WORK

The paper "A Tutorial on Principal Component Analysis" by Jonathon Shlens explains PCA as a method for simplifying complex data by identifying key patterns. [1] It provides both intuitive examples and rigorous mathematical derivations, emphasizing its assumptions and practical applications in data analysis. Similarly, The paper 'A Comparison of Eigenvalue Methods For Principal Component Analysis' compares four eigenvector methods for PCA-based face recognition: cyclic Jacobi, Wiedlandt's deflation, Hotelling's deflation, and MATLAB's eigenvalue method.[2] MATLAB's method is fastest and scales well, while Hotelling's offers modifiable code. Filtering face images can improve recognition rates.

with working with high-dimensional data, includes serious problems like data sparsity, which makes it difficult to draw conclusions that are meaningful, and increased similarity in distances between data points, which makes it more difficult to distinguish between similarities and differences. Additionally, higher dimensions demand more data to bridge the gap created by added dimensions, leading to heightened computational demands and a greater risk of overfitting. PCA offers a systematic approach to address these challenges,

III. METHODOLOGY

A. THEORY

Principal Component Analysis (PCA) aims for reducing data dimensionality by resizing the dimensions to a desired number, aiming for the most accurate representation possible. The "curse of dimensionality," or the difficulties associated

extracting vital information while discarding irrelevant details through a step-by-step process. By determining the appropriate number of components, PCA constructs a vector representing the data in orthogonal axes, effectively reducing dimensionality while preserving essential characteristics. The standardization process typically involves subtracting the mean from the entire dataset, possibly with division by standard deviation, ensuring uniformity in data representation. This methodology extends to both manually generated datasets and standard datasets, involving processes like fit transform to ensure consistency in analysis. Through these techniques, PCA offers a viable solution to the curse of dimensionality, enabling efficient data analysis and interpretation.

B. MATHEMATICAL FORMULAE

1. Generating a random projection matrix

The projection matrix used for standardization is made by selecting elements from a Standard Normal Distribution. The normal distribution follows the formula:

$$F(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma} \right)^2} \quad (1)$$

2. Transformation of datasets

Transformation of the synthetic dataset is done by multiplying the Randomly Generated Matrix of size 20x2 by the 2x2 matrix generated by selecting elements from a standard normal distribution.

$$\text{Transformed Data} = \begin{bmatrix} a_{11} & b_{12} & \dots & a_{1n} \\ a_{21} & b_{22} & \dots & a_{2n} \end{bmatrix} \begin{bmatrix} c_1 & d_1 \\ c_2 & d_2 \end{bmatrix} \quad (2)$$

The standard dataset is passed through the fit-transform, which works as below:

$$\text{Transformed Data} = \left(\frac{x - \mu}{\sigma} \right) \quad (3)$$

3. Calculation of Variance along dominant axis

$$\text{Variance} = \frac{\sum (x - \text{mean})^2}{N} \quad (4)$$

4. Covariance Matrix

$$S_x = \frac{1}{n-1} (X^T \cdot X) \quad (5)$$

5. Eigenvectors and eigenvalues

For a given square matrix B, the equation for eigenvectors and eigenvalues is:

$$B \cdot v = \lambda v \quad (6)$$

In Equation 6, v represents an eigenvector and λ is the corresponding eigenvalue. To find the eigenvalues, we rearrange the equation as:

$$(B - \lambda_i I) \cdot v_i = 0 \quad (7)$$

6. Proportion of Variance

The proportion of variance for each of the eigenvalue can be calculated as:

$$\text{Proportion of Variance} = \frac{\lambda_i}{\sum_{i=1}^n \lambda_i} \quad (8)$$

C. SYSTEM BLOCK DIAGRAM

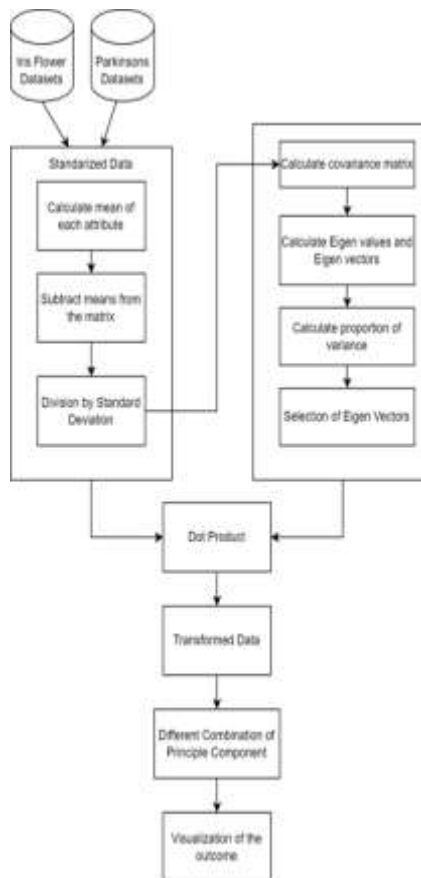


Figure 1. System Block Diagram

D. INSTRUMENTATION TOOLS

The process is entirely done using Python. Jupyter Notebook, a popular interactive coding environment was used for the experimentation. It provides several essential libraries for data processing and visualization were used in the code. To effectively manage arrays and perform numerical calculations, the numpy library was imported. A popular dataset for machine learning applications, the Iris dataset, was imported using the sklearn.datasets module. Plots and visualisations of all kinds could be made with the matplotlib.pyplot library. Plotting in three dimensions was made possible by the import of the mpl_toolkits.mplot3d module. Lastly, combinations were created for certain analysis tasks using the Itertools library. In the Jupyter Notebook environment, these libraries were essential for carrying out data analysis and enabling thorough experimentation.

E. WORKING PRINCIPLE

1. Dataset Preparation

The UC Irvine Machine Learning Repository is the source of two common datasets. The library is used to import the iris and Parkinsons Classification datasets. After that, the data are normalized to have a zero mean and a one standard deviation by applying the fit transform function. Using numpy libraries, a random dataset is created. Data points from a conventional normal distribution are then used to create a projection matrix, which transforms the data.

2) Eigenvector Computation

The covariance of the datasets is then calculated. Eigenvalues and eigenvectors are derived from the covariance matrix. The Eigen vectors are chosen as the primary components for analysis, and the proportion of variance is also computed. The dataset is analyzed and transformed using the dot product between the chosen component and the standardized dataset.

3) Data Analysis and Visualization

The covariance matrix, which contains the PCA result, is computed using the modified data. Several Python data visualization libraries are then used to visualize the result. The procedure is repeated using several main component combinations chosen for data analysis and visualization.

IV. RESULTS

A. PCA ON RANDOM DATASET

We first created a random dataset and performed PCA from scratch. We created an array of 20 by 2 samples by taking samples from a univariate "normal" (Gaussian) distribution, and then we produced the scatter plot. This allowed us to understand how PCA helps in dimensionality reduction and enabled us to visualize the data prior to using PCA. We could see how well PCA captured variation and reduced the

dataset's dimensionality by projecting the data onto the main components.

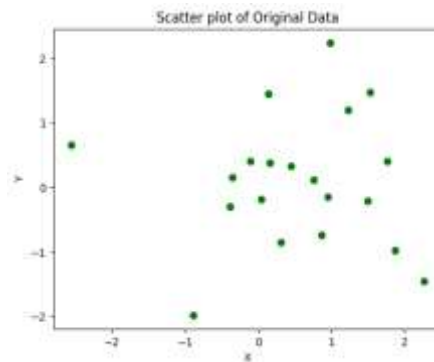


Figure 2. Scatter plot of Normal (Gaussian) data

Figure 2 shows the scatter plot of the resultant matrix that we obtained by multiplying the original matrix with a 2 by 2 matrix with random values from a uniform distribution having values between 0 and 1.

We then calculated the variance along each axis of the transformed data and computed the covariance matrix of the transformed data. The covariance matrix captures the pairwise covariance between the dimensions, indicating how much the dimensions vary together. We determined the eigenvalues and eigenvectors of the covariance matrix using which we performed a change of basis to transform the data into a new coordinate system aligned with the principal components.

This scatter plot represents how the data is represented in the new basis.

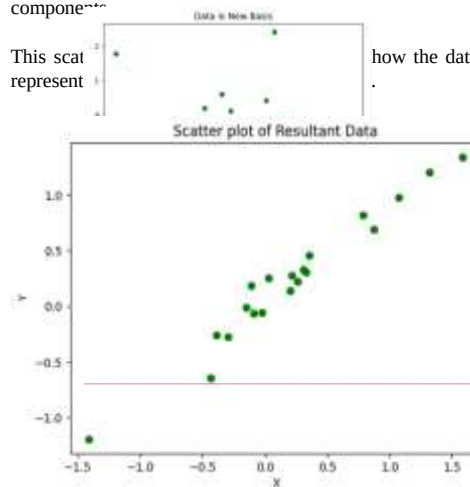


Figure 3. Plot of Resultant data

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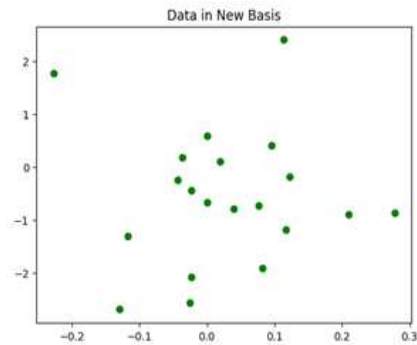


Figure 4. Plot of data in new basis

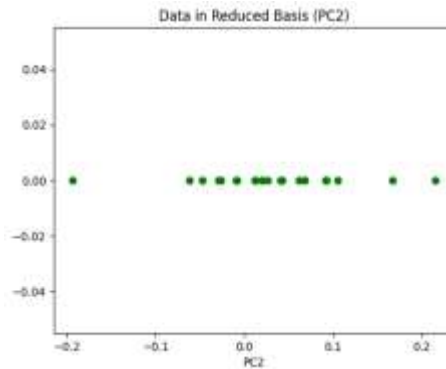


Figure 7. One Dimension Plot using PC2

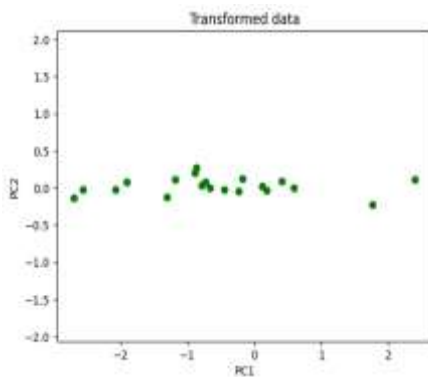


Figure 5. Plot of transformed data

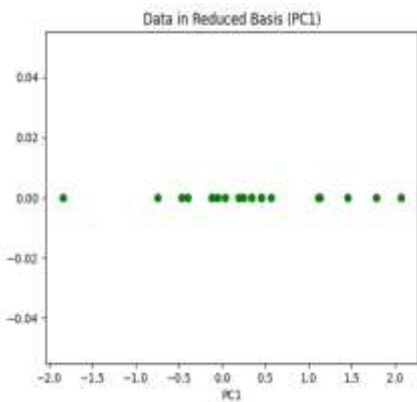


Figure 6. One Dimension plot using PC1

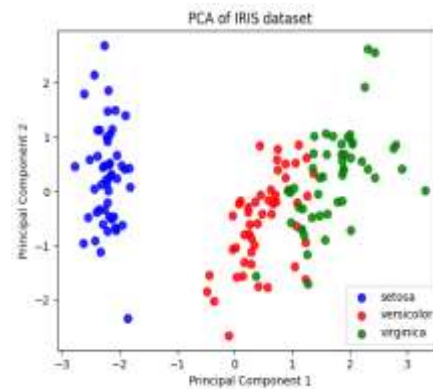


Figure 8. PC1 vs PC2 of Iris Dataset

B. PCA ON IRIS DATASET

We performed PCA on the Iris dataset to minimize its dimensionality while retaining the greatest variance. The dataset, which included 150 samples with four variables each (sepal length, sepal width, petal length, and petal width), was standardized so that each feature had a mean of 0 and a standard deviation of 1. PCA was then used to reduce the dataset to four major components. We used 2D scatter plots to display the results between different pairs of principal components, as well as a 3D plot of the first three principle components. These visualizations serve to show the structure of the data as well as the variance captured by the principal components.

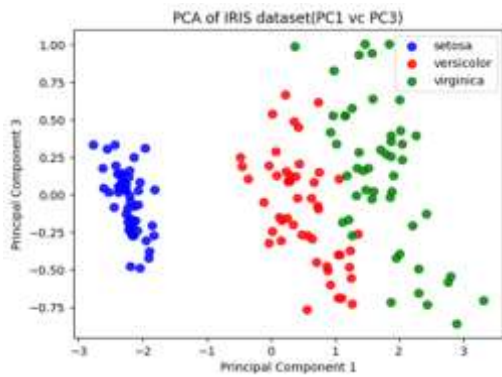


Figure 9. PC1 vs PC3 of Iris Dataset

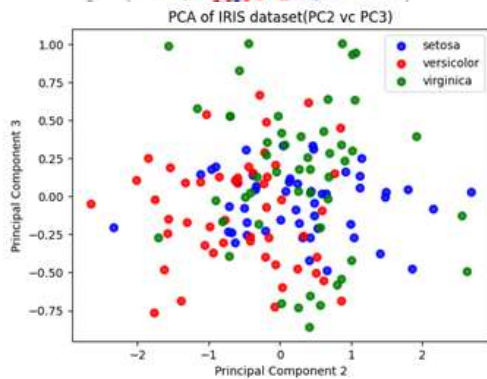


Figure 10. PC2 vs PC3 of Iris Dataset

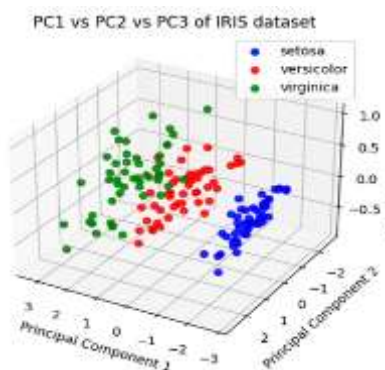


Figure 11. 3D plot of PC1 vs PC2 vs PC3

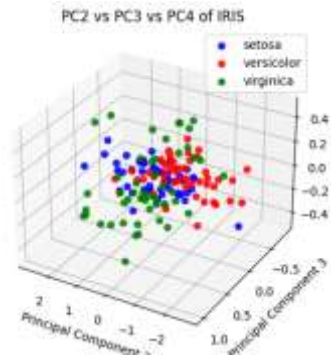


Figure 12. 3D plot of PC2 vs PC3 vs PC4

C. PCA ON PARKINSONS DATASET

We performed PCA on Oxford Parkinson's Disease Detection Dataset obtained from UC Irvine Machine Learning Repository. The dataset contains a total of 195 instances, each with 22 attributes. These attributes include various biomedical voice measurements, which can be used to predict the presence of Parkinson's disease. This helps in visualizing patterns and identifying potential clusters in the data, which may correspond to different health statuses.

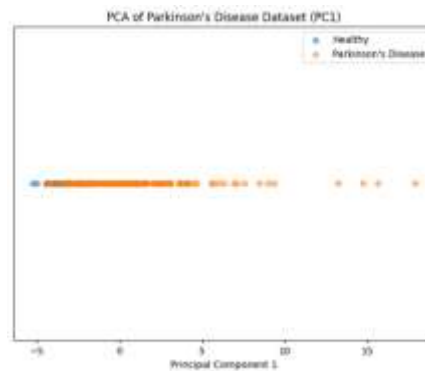


Figure 13. Plot of PC1 on Parkinson's dataset

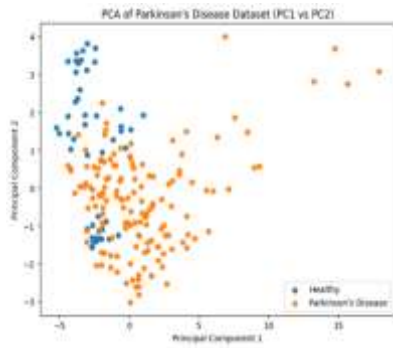


Figure 14. PC1 and PC2 in Parkinson's dataset

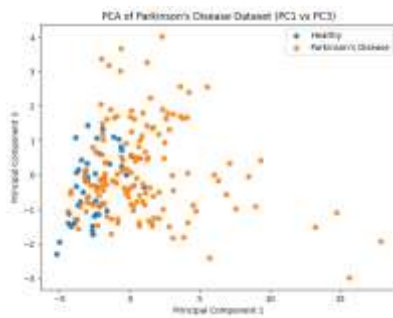


Figure 15. PC1 and PC3 in Parkinson's dataset

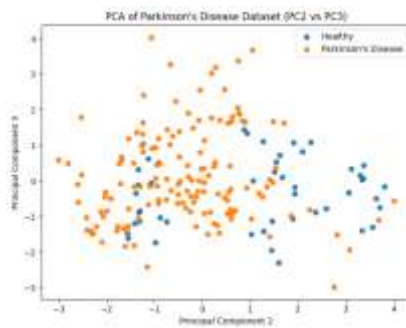


Figure 16. PC2 and PC3 in Parkinson's dataset

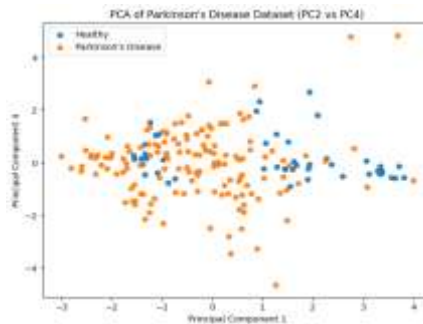


Figure 17. PC2 and PC4 in Parkinson's dataset

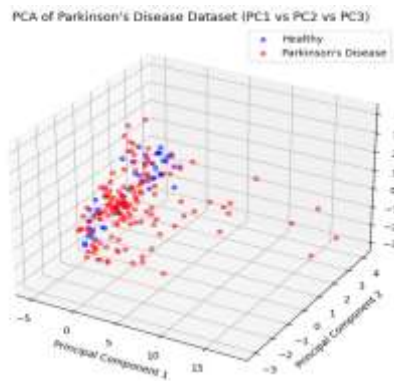


Figure 18. 3D plot of PC1, PC2 and PC3 in Parkinson's dataset

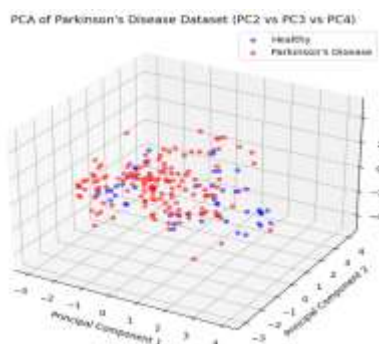


Figure 19. 3D plot of PC2, PC3 and PC4 in Parkinson's dataset

The first four principal components have the following explained variance ratios:

- PC1: 0.589 (58.9%)
- PC2: 0.113 (11.3%)
- PC3: 0.070 (7.0%)
- PC4: 0.067 (6.7%)

The explained variance ratio represents the proportion of the dataset's variance captured by each principal component.

These values indicate that the first principal component (PC1) captures approximately 58.9% of the total variance in the dataset. PC2 captures an additional 11.3% of the variance, followed by PC3 and PC4, which capture 7.0% and 6.7% of the variance, respectively. Together, these four components account for approximately 83.9% of the total variance in the dataset.

The scree plot displays the eigenvalues (which correspond to the amount of variance captured) of each principal component on the y-axis, with the principal component number on the x-axis.

In this case, the scree plot reveals that PC1 contributes the most significant amount of variance. Hence, selecting the first few principal components for further analysis is often sufficient to capture the most important features of the data while reducing dimensionality.

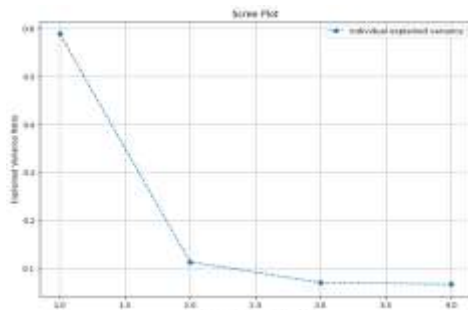


Figure 20. Scree plot

V. DISCUSSION AND ANALYSIS

The preceding sections demonstrated the effective use of Principal Component Analysis (PCA) to reduce dimensionality. PCA was applied to the Parkinson's Disease dataset, and the findings were presented in a variety of graphs. This approach drastically decreased the dataset's complexity, making it easier to maintain and comprehend.

This reduction is crucial for enhancing data manageability and comprehension, especially when dealing with large datasets that have multiple features. The findings were presented through various graphs, including 2D and 3D scatter plots, which clearly illustrated the distribution of data points in the new principal component space. These visualizations provided insights into the underlying structure of the data, revealing patterns and relationships

that were not easily observable in the original high-dimensional space.

When the patterns revealed by several pairs of principle components were examined, it was discovered that the scatter plot of the first two principal components (PC1 and PC2) displayed a clearer and more significant pattern. This is because PC1 and PC2 capture the greatest amount of variance in the data, maintaining the most important structure and relationships. However, the scatter plot of the second and third principal components (PC2 and PC3) did not disclose a distinct trend. This is predicted given that PC3 captures less variance than PC1 and PC2, implying that it provides less relevant structural information in the dataset. The diminished clarity of the PC2 vs. PC3 figure implies that PC3 catches more noise or less significant fluctuations in the data.

The transformation resulted in a covariance matrix with fewer off-diagonal items and stronger diagonal elements, showing that PCA successfully concentrated the dataset's variability in fewer dimensions. This result is consistent with PCA's core goal of maximizing variance in a reduced dimensional space.

VI. CONCLUSION

To sum up, this lab followed the guidelines for Principal Component Analysis (PCA) effectively. We tested PCA on different datasets and found it worked well in simplifying them. By reducing the number of components, PCA helped us focus on the most important parts of the data, making it easier to understand. Our results showed that using more components tended to improve PCA's performance. However, we also discovered some limitations of PCA. For instance, it struggled when there were outliers in the data, which could distort our analysis. Additionally, when we used fewer components, it became harder to distinguish patterns in the data. Moreover, PCA assumes that the relationships in the data are linear, which may not always be the case. Despite these challenges, PCA remains a valuable tool for data analysis. It helps us uncover insights and simplify complex datasets. However, it's important to be mindful of its limitations and consider them when interpreting the results.

REFERENCES

- [1] J. Shlens, "A Tutorial on Principal Component Analysis," 2014.
- [2] Y. Danisman, M. F. Yilmaz, A. Ozkaya and İ. T. Çömlekçiler, "A comparison of eigenvalue methods for principal component analysis," 2014.



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